

Notes: Fleckenstein & Longstaff (RFS, 2025)

Do Municipal Bond Investors Pay a Convenience Premium to Avoid Taxes?

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1 Big picture

- **Question.** Do investors pay extra for tax-exempt municipal bonds because those bonds provide a form of tax-related convenience, beyond the cash-flow value implied by credit risk, liquidity, and the maximum possible tax benefit?
- **Main idea.** Treasury investors pay convenience premia for safety and liquidity. This paper asks whether municipal investors also pay a convenience premium for the ability to shelter wealth from taxation.
- **Market setting.** The paper studies state-issued general obligation municipal bonds, whose repayment source is the state general fund.
- **Identification.** Credit is measured using state CDS curves. Liquidity is identified using taxable municipal bonds because taxable munis and Treasuries receive similar tax treatment.
- **Main magnitude.** The average estimated convenience premium is 15.49 basis points across the 2008–2021 sample.
- **Tax-aversion evidence.** Premia are related to tax uncertainty, municipal fund flows, migration to low-tax states, tax-minimizing strategies, and the TCJA SALT deduction shock.
- **Economic takeaway.** Investors appear to value freedom from taxation itself, creating a convenience premium and a form of tax-related seigniorage for municipal issuers.

2 Data and measurement

2.1 Municipal bond sample

The sample focuses on state-issued general obligation bonds for which repayment comes from the state’s general fund. The authors manually collect bond characteristics from Bloomberg and transaction prices from MSRB through WRDS. The sample excludes callable, puttable, insured, prerefunded, floating-rate, AMT, OID, discount, 144A, and otherwise nonstandard bonds.

Interpretation. The narrow sample makes the credit and liquidity decomposition cleaner.

2.2 Benchmarks and CDS

Each municipal bond is matched to a synthetic Treasury and to a repo-OIS riskless benchmark with the same coupon and maturity. State CDS data for 21 states are used to remove issuer default risk.

Interpretation. State general obligation bonds plus state CDS provide a cleaner credit-risk adjustment than treating credit as a residual.

2.3 Tax treatment

Tax-exempt municipal interest is exempt from federal income taxation and generally from state taxes for in-state residents. Taxable municipal bonds and Treasuries are both federally taxable and generally state-tax exempt for the relevant investor. Repo OIS and CDS cash flows are treated as ordinary income.

Interpretation. Taxable munis are the key asset for identifying municipal-bond liquidity.

3 Models and empirical specifications

The adjusted yield removes credit and liquidity:

$$y_{i,t}^{Adj} = y_{i,t}^{obs} - CreditSpread_{i,t} - LiquiditySpread_{i,t}.$$

The liquidity spread is estimated from taxable municipal bonds:

$$y_{i,t}^{Adj,TaxableMuni} - y_{i,t}^{Adj,Treasury} = \alpha_t + \beta_1 Age_{i,t} + \beta_2 Maturity_{i,t} + \beta_3 TradeSize_{i,t} \\ + \beta_4 TradingVolume_{i,t} + \beta_5 AmountIssued_i + \beta_6 CDS_{i,t} + \varepsilon_{i,t}.$$

The convenience premium is defined by

$$CP = r_{Tax}(1 - \tau_{max}) - r_{TaxEx},$$

where r_{TaxEx} is the adjusted tax-exempt municipal yield, r_{Tax} is the pretax matched taxable benchmark yield, and τ_{max} is the maximum combined marginal federal, state, and NIIT tax rate.

Interpretation. A positive value means the tax-exempt yield is below even the lowest possible after-tax taxable benchmark yield.

4 Identification and empirical design

- State CDS identify credit risk because the bonds are state-issued general obligations.
- Taxable munis identify liquidity because they are taxed similarly to Treasuries.
- Using the maximum possible marginal tax rate makes the convenience-premium estimate conservative.
- Tax-aversion evidence comes from tax uncertainty, state muni fund flows, migration, IRA and retirement contributions, charitable contributions, itemized deductions, and the TCJA SALT limitation.
- The main limitations are the restriction to states with CDS, assumptions about CDS price discovery, and the assumption that taxable and tax-exempt municipal liquidity can be captured in one model.

5 Tables and figures

The visuals below are directly cropped from the original paper and grouped to save space.

5.1 Figure 1 + Table 1: identifying the liquidity component

Read:

- Figure 1 shows the monthly spread between credit-adjusted taxable municipal yields and matched Treasury yields.
- Table 1 estimates the liquidity model: age and maturity raise liquidity spreads; trade size and trading volume lower them.

Economic message. The taxable muni–Treasury spread provides the paper’s liquidity adjustment.

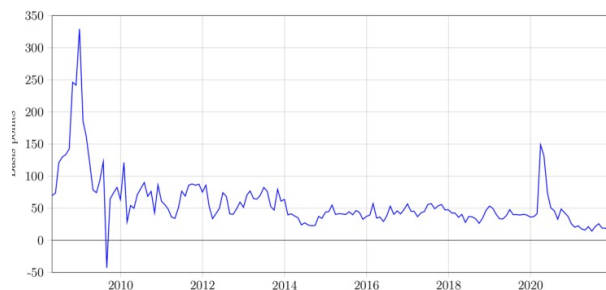


Figure 1
Credit-adjusted yield spreads between taxable municipal bonds and Treasuries
The graph shows the monthly average spread between the credit-adjusted yield of taxable municipal bonds and

(a) Figure 1

Table 1
The liquidity model

Explanatory variable	Coeff	t-stat
Age	5.4069	6.88**
Maturity	4.5357	6.89**
Trade size	-0.5980	-2.25**
Trading volume	-0.0239	-2.90**
Amount issued	0.0016	0.24
CDS spread	-0.1186	-3.16**
Monthly fixed effects		Yes
Adj. R^2		0.5179
N		13,331

This table reports the results from the regression used to parameterize the liquidity model. The dependent variable in the regression is the difference between the credit-risk-adjusted yield of a taxable state-issued bond and the credit-risk-adjusted yield of the equivalent matched-coupon-and-maturity Treasury security. Age and Maturity are measured in years. Trade size is the average notional value per trade for the taxable municipal bond during the month of the observation and expressed in millions of dollars. Trading volume is the total notional value of the taxable municipal bond traded during the month of the observation and expressed in millions of dollars. Amount issued is the total notional amount of the taxable bond issue and expressed in millions of dollars. CDS spread is the CDS spread for the issuing state as of the end of the month of the observation and expressed in basis points. Standard errors are based on Liang and Zeger (1986) and clustered by state and year. The sample period is from April 1, 2008 to December 30, 2021. * $p < .1$; ** $p < .05$.

(b) Table 1

5.2 Table 2 + Table 3: yield components and tax pricing

Read:

- Across all states, the average yield is 1.519%, credit is 0.370%, liquidity is 0.600%, and the adjusted yield is 0.548%.
- Table 3 shows that higher marginal tax rates predict richer tax-exempt bond prices.

Economic message. Credit and liquidity are large components of muni yields, but tax shelter remains priced after removing them.

Table 2
Summary statistics for the components of tax-exempt state-issued municipal bond yields

State	Yield	Credit	Liquidity	Adj. yield	N
California	1.644	0.506	0.600	0.538	133,102
Connecticut	1.658	0.485	0.518	0.655	47,128
Delaware	1.364	0.223	0.600	0.541	11,873
Florida	1.473	0.278	0.542	0.653	26,219
Maryland	1.358	0.275	0.638	0.446	44,530
Massachusetts	1.533	0.377	0.716	0.440	60,368
Michigan	1.764	0.476	0.733	0.555	5,352
Minnesota	1.301	0.225	0.564	0.512	30,305
Mississippi	1.396	0.217	0.572	0.607	2,469
Nevada	1.613	0.358	0.565	0.690	9,209
New Jersey	1.802	0.617	0.658	0.528	13,130
New York	1.453	0.449	0.685	0.319	10,645
Ohio	1.532	0.367	0.627	0.537	59,062
Pennsylvania	1.523	0.379	0.554	0.589	35,005
Rhode Island	1.514	0.319	0.498	0.697	4,187
South Carolina	1.352	0.196	0.544	0.612	6,924
Texas	1.522	0.277	0.655	0.590	27,872
Utah	1.039	0.242	0.524	0.273	7,546
Virginia	1.300	0.282	0.625	0.393	6,148
Washington	1.417	0.236	0.545	0.637	48,743
Wisconsin	1.441	0.252	0.534	0.655	26,797
All	1.519	0.370	0.600	0.548	616,614

This table reports summary statistics for the decomposition of the tax-exempt state-issued municipal bond yields into separate components for the indicated states. Yield denotes the average observed bond yield. Credit denotes the average value of the credit-related component of the bond yield. Liquidity denotes the average value of the liquidity-related component of the bond yield. Adj. yield is the average credit-and-liquidity-adjusted bond yield and is calculated by subtracting the sum of the credit and liquidity components of the yield from the observed bond yield. Yield, credit, liquidity, and N denote the number of observations. The sample period is from April 1, 2008 to December 30, 2021.

(a) Table 2

Table 3
Cross-sectional test for convenience premia

Explanatory variable	Coeff	t-stat
Difference in marginal rates	0.7795	3.51**
Monthly fixed effects		Yes
Adj. R^2		0.0825
N		234,925

This table reports the results from the panel regression of the difference between the adjusted yields of pairs of tax-exempt bonds from different states, where the bonds in the pairs have the same coupon rate and maturity dates within 3 months of each other, on the difference in the top marginal tax rates for those states. The difference in the adjusted yields is expressed in basis points. The difference in the top marginal tax rates is expressed as a percentage. Standard errors are based on Liang and Zeger (1986) and clustered by year. The sample period is from April 1, 2008 to December 30, 2021. * $p < .1$; ** $p < .05$.

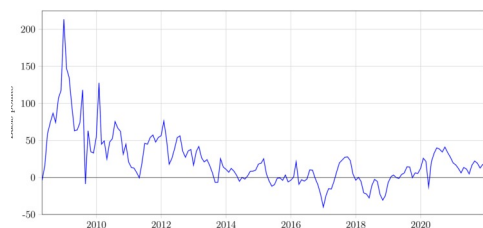
(b) Table 3

5.3 Figure 2 + Table 4: estimating the convenience premium

Read:

- Figure 2 plots the monthly average convenience premium.
- Table 4 reports state-year premia; the all-state/all-year average is 15.49 basis points.

Economic message. Tax-exempt munis are often richer than can be explained by credit, liquidity, and the maximum mechanical tax benefit.



(a) Figure 2

Table 4
Summary statistics for the convenience premia

State	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	Mean
California	80.43	42.73	37.77	12.98	35.33	14.52	0.57	4.84	-0.11	8.46	-16.98	-0.41	28.55	13.50	14.76
Connecticut	-	77.06	60.15	51.27	44.31	16.87	8.38	2.29	-15.36	-28.21	-43.02	-10.91	-1.05	16.29	4.25
Delaware	-	47.70	39.57	32.52	34.16	21.05	10.60	6.60	-5.47	8.54	-11.84	8.92	24.08	12.85	15.33
Florida	94.04	67.57	43.81	20.49	31.05	7.93	7.26	-0.19	-0.16	8.45	-1.18	13.71	20.17	11.08	11.02
Maryland	108.47	113.40	54.82	48.48	55.83	32.59	15.97	3.89	-1.40	11.71	-22.04	1.91	30.32	10.92	25.35
Massachusetts	124.20	98.32	66.16	45.96	53.66	31.65	19.22	13.71	0.63	15.82	-6.55	13.18	34.74	22.49	32.47
Michigan	117.71	71.25	64.31	18.59	44.51	24.64	15.05	2.23	-0.71	4.24	-16.68	5.31	25.32	13.10	24.01
Minnesota	-	26.83	34.63	36.79	21.82	6.86	-	-	-5.82	6.93	-19.33	0.59	22.24	7.72	10.39
Mississippi	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nevada	-	39.67	46.39	1.70	10.74	0.53	-8.56	-4.60	-7.52	8.42	1.84	17.18	20.96	19.75	8.14
New Jersey	129.18	106.93	70.98	29.93	51.92	27.52	14.72	-10.49	-7.62	-10.58	-29.86	-6.16	-10.83	-6.42	18.35
New York	102.26	98.21	78.79	35.30	44.91	12.15	6.00	6.74	3.91	18.76	-16.36	6.69	18.86	11.75	32.80
Ohio	113.26	82.15	51.67	31.54	34.89	10.31	7.77	0.38	-3.98	8.78	-13.77	7.38	26.51	15.21	17.55
Pennsylvania	-	-	48.29	35.24	43.07	13.39	0.83	-6.08	-15.77	-5.21	-25.58	2.21	20.53	20.74	6.53
Rhode Island	-	-	27.94	2.46	5.23	3.09	-16.56	-17.63	-17.83	-1.13	-14.16	5.85	3.50	15.86	-1.09
South Carolina	-	-	-	40.62	16.52	3.18	-	-0.45	-7.50	4.49	-13.31	9.20	23.63	17.63	5.69
Texas	108.30	78.54	43.58	30.78	33.45	17.21	11.30	-1.63	-6.59	11.05	1.01	15.49	22.29	14.70	20.97
Utah	-	-	-	-	38.78	37.28	33.38	23.81	10.98	16.14	-14.69	7.18	23.35	9.28	20.98
Virginia	-	95.28	68.01	54.14	43.13	28.85	13.08	4.45	0.33	14.94	-6.55	17.63	23.56	17.09	31.21
Washington	-	-	-	25.16	22.86	15.76	-0.01	-6.83	-13.20	7.46	0.29	15.56	23.49	10.69	8.18
Wisconsin	-	-	15.30	19.21	24.69	2.02	-5.72	-11.18	-14.92	8.07	-16.88	1.54	18.53	11.21	1.72
Mean	99.32	74.15	51.72	32.08	38.46	17.39	5.75	0.86	-6.48	4.28	-15.20	4.98	23.14	13.91	15.49

This table reports the annual average values of the convenience premia of the state-issued tax-exempt municipal bonds for the indicated states and years. The last column reports the averages taken across all years by state. The last row reports the averages taken across all states by year. The value in the last row and column is the value across all states and years. The convenience premia are expressed in basis points. The sample period is from April 1, 2008 to December 30, 2021.

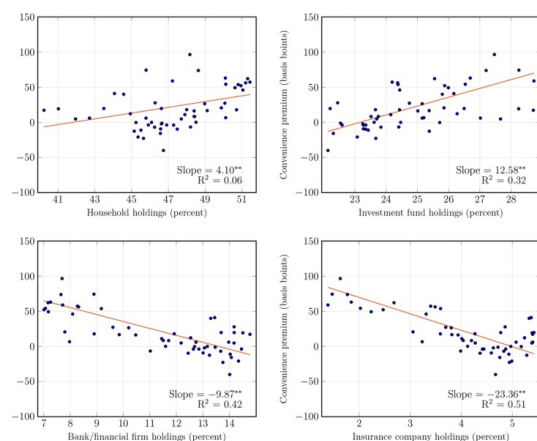
(b) Table 4

5.4 Figure 3 + Tables 5–6: ownership, other premia, and uncertainty

Read:

- Figure 3 links premia to household/investment-fund ownership and away from bank/insurance ownership.
- Table 5 links municipal convenience premia to other fixed-income convenience premia.
- Table 6 links premia to VIX, MOVE, tax uncertainty, and government-spending uncertainty.

Economic message. Municipal premia have a broad safe-asset component and a unique tax-related component.



(a) Figure 3

Table 5
Regression of the convenience premium in the tax-exempt municipal bond market on convenience premium measures from other markets

Explanatory variable	Coeff	t-stat
Intercept	-63.6647	-7.07**
Repo spread	-0.1800	-1.25
TED spread	-0.4145	-4.67**
Swap spread	0.3958	2.01**
AAA spread	0.1970	3.09**
Treasury richness	0.4008	2.23**
Refcorp spread	0.2341	7.37**
Agency MBS spread	0.3238	2.48**
Adj. R²	0.6944	
N	165	

This table reports the results from the regression of the monthly average convenience premium in the tax-exempt municipal bond market on convenience premium measures from other markets. The monthly average convenience premium is the simple average taken across all convenience premium observations for all states each month. Repo spread is the difference between the 3-month general collateral repo rate and the 3-month Treasury bill rate. TED spread is the difference between the 3-month U.S. LIBOR rate and the 3-month Treasury bill rate. Swap spread is the 2-year LIBOR-based swap spread. AAA spread is the difference between the yield of the Bloomberg index of AAA-rated corporate bonds and the yield on 10-year Treasuries. Treasury richness is the Fitch/IHS and Longstaff (2020) Treasury richness measure for Treasuries with maturities less than or equal to 10 years. Refcorp spread is the average difference between the yields of Refcorp bonds with maturities less than or equal to 10 years and matched-maturity Treasuries. Agency MBS spread is the option-adjusted yield spread between Agency MBS and Treasuries. All variables are expressed in basis points. Standard errors are

Table 6
Regression of the convenience premium on risk and uncertainty measures

Explanatory variable	Coeff	t-stat
Intercept	-39.8486	-4.69**
VIX index	0.9871	1.81*
MOVE index	0.4983	3.73**
Tax uncertainty	0.1427	2.84**
Gov spending uncertainty	0.0315	1.87*
Economic uncertainty	-0.1220	-2.61**
Adj. R²	0.5546	
N	165	

This table reports the results from the regression of the monthly average convenience premium in the tax-exempt municipal bond market on the indicated variables. The monthly average convenience premium is the simple average taken across all observations for all states each month and is measured in basis points. VIX index is the Chicago Board Options Exchange (CBOE) Index of S&P 500 option-implied volatility. MOVE index is the Intercontinental Exchange's U.S. Treasury volatility index. Tax uncertainty, Gov spending uncertainty, and Economic uncertainty are the Baker, Bloom, and Davis (2016) indices of tax uncertainty, government spending uncertainty, and economic policy uncertainty, respectively. Standard errors are based on Newey and West (1987).

(b) Table 5

(c) Table 6

5.5 Tables 7–10: fund flows, migration, tax strategies, and SALT

Read:

- Table 7 shows that premium changes forecast flows into single-state municipal bond funds.

- Table 8 links premia to outmigration to no-income-tax states.
- Table 9 links premia to IRA, retirement, charitable, and itemized deduction measures.
- Table 10 uses the TCJA SALT limitation shock.

Economic message. These tests connect the premium to revealed tax aversion.

Table 7
Results from the convenience premium and state municipal bond funds net fund flows regressions

Explanatory variable	Net flow regression		Δ Premium regression	
	Coeff	<i>t</i> -stat	Coeff	<i>t</i> -stat
<i>Net flow</i> _{<i>t</i>-1}	0.4027	3.18**	0.0002	0.03
<i>Net flow</i> _{<i>t</i>-2}	0.0137	0.12	-0.0238	-2.97**
<i>Net flow</i> _{<i>t</i>-3}	0.0967	1.76*	0.0126	2.23**
Δ <i>Premium</i> _{<i>t</i>-1}	0.2361	2.26**	-0.2806	-7.03**
Δ <i>Premium</i> _{<i>t</i>-2}	0.2393	2.47**	-0.0899	-2.92**
Δ <i>Premium</i> _{<i>t</i>-3}	0.2316	2.87**	-0.1673	-6.28**
State fixed effects	Yes		Yes	
Adj. <i>R</i> ²	0.2397		0.1043	
<i>N</i>	2,221		2,221	

The first column reports the results from the panel regressions of net flows into individual state municipal bond funds on the indicated lagged changes in the average monthly convenience premium for the corresponding state. The average monthly convenience premium for each state is the simple average taken across all observations for that state each month and is measured in basis points. A state municipal bond fund is an open-end mutual fund that invests in the tax-exempt municipal bonds of that state. The states in the sample for which there are state municipal bond funds are California, Connecticut, Maryland, Massachusetts, Michigan, Minnesota, Mississippi, New Jersey, New York, Ohio, Pennsylvania, Rhode Island, South Carolina, Utah, Virginia, and Wisconsin. The second column reports the results from the panel regressions of changes in the average monthly convenience premium for individual states on the indicated lagged net flows into state municipal bond funds for the corresponding state. The convenience premium is expressed in basis points. Net fund flows are expressed in millions of dollars. Standard errors are based on White (1980). The sample period is monthly from April 2008

(a) Table 7

Table 8
Regression of changes in the convenience premia on state-level migration flows

Explanatory variable	Coeff	<i>t</i> -stat
<i>Migration</i>	102.2761	2.03**
State fixed effects	Yes	
Adj. <i>R</i> ²	0.0460	
<i>N</i>	199	

This table reports the results from the panel regression of year-end to year-end changes in the monthly convenience premium for individual states on the percentage of that state's population that moved to one of the seven states without a state income tax. The monthly convenience premium for each state is the simple average taken across all observations for that state that month and is measured in basis points. The states without a state income tax are Alaska, Florida, Nevada, South Dakota, Texas, Washington, and Wyoming. Convenience premia are expressed in basis points. Standard errors are based on White (1980). The sample period is annual from 2008 to 2019. **p* < .1; ***p* < .05.

(b) Table 8

Table 9
Regressions of the convenience premium on tax-minimizing strategies

Explanatory variable	Coeff	<i>t</i> -stat	Coeff	<i>t</i> -stat	Coeff	<i>t</i> -stat	Coeff	<i>t</i> -stat
<i>IRA contribution</i>	47.2107	3.24**	—	—	—	—	—	—
<i>Tax-deferred retirement</i>	—	—	104.2937	2.19**	—	—	—	—
<i>Charitable contribution</i>	—	—	—	—	1.9454	4.10**	—	—
<i>Itemized deductions</i>	—	—	—	—	—	—	1.4039	3.06**
State fixed effects	Yes		Yes		Yes		Yes	
Year fixed effects	Yes		Yes		Yes		Yes	
Adj. <i>R</i> ²	0.8360		0.9399		0.9421		0.9407	
<i>N</i>	232		241		241		241	

This table reports the results from the regressions of the year-end convenience premia for individual states on the percentage of tax returns for each state that include an IRA contribution, a self-employed retirement plan, itemized deductions, or charitable contributions on Schedule A. The year-end convenience premia for each state are the monthly convenience premia for the month of December. The monthly convenience premium for each state is the simple average taken across all observations for that state that month and is measured in basis points. Data are from the Internal Revenue Service Statistics of Income reports. Standard errors are based on White (1980). The sample period is annual from 2008 to 2020. **p* < .1; ***p* < .05.

(a) Table 9

Table 10
SALT-limitation event study

Explanatory variable	Coeff	<i>t</i> -stat
Intercept	-37.4890	-10.16**
Δ <i>SALT deduction</i>	32.6418	2.57**
Adj. <i>R</i> ²	0.2449	
<i>N</i>	21	

This table reports the results from the cross-sectional regression of the change in the monthly convenience premium from September 2017 to March 2018 for the individual states on the percentage change in the state and local tax (SALT) deduction per income tax return for that state between the 2017 and 2018 tax years. The monthly convenience premium for each state is the simple average taken across all observations for that state that month and is measured in basis points. Data are from the Internal Revenue Service Statistics of Income reports. Standard errors are based on White (1980). **p* < .1; ***p* < .05.

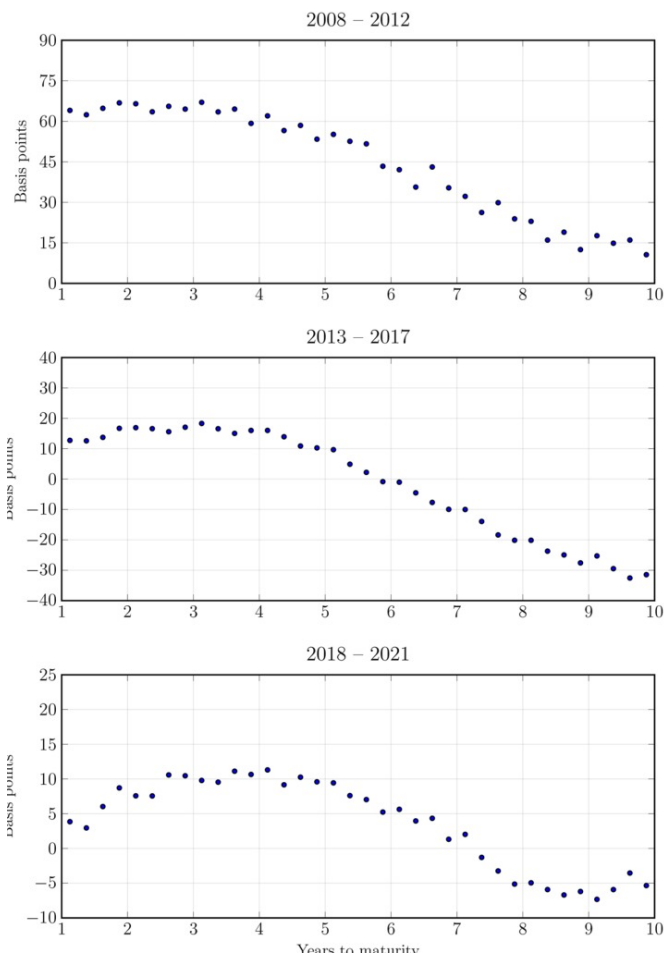
(b) Table 10

5.6 Figure 4 + Table 11: convenience premium versus tax-risk premium

Read:

- Figure 4 shows that premia generally decline with maturity.
- Table 11 shows that implied forward tax rates frequently exceed 100%, especially at short maturities.

Economic message. Tax risk may explain part of the premium, but cannot explain all of it.



(a) Figure 4

Table 11
Percentage of implied forward tax rates exceeding 100%

Maturity range	Implied forward tax rate > 100%
1–2 years	55.37
2–3 years	45.98
3–4 years	36.89
4–5 years	26.88
5–6 years	18.86
6–7 years	11.72
7–8 years	6.80
8–9 years	4.28
9–10 years	3.50

This table reports the percentage of implied forward tax rates greater than 100% for the indicated maturity categories. Implied forward tax rates are calculated by solving for the forward tax rate implied by the difference between the adjusted yield of the tax-exempt municipal bond and the yield of the corresponding taxable benchmark bond and using the current maximum combined federal and state tax rates over the next year. Implied forward tax rates are annualized and expressed as a percentage. The column Implied forward tax rate > 100% presents the percentage of observations where the implied forward tax rate is greater than 100% for municipal bonds with remaining maturities in the indicated maturity ranges. The sample period is from April 1, 2008 to December 30, 2021.

(b) Table 11

5.7 Table 12: issuance and convenience premia

Read:

- Unexpected issuance lowers subsequent premia.
- Higher premia predict subsequent issuance.

Economic message. Issuers respond to tax-related richness by issuing more debt.

Table 12**Results from the convenience premium and unexpected municipal bond issuance regressions***A. Convenience premium results*

Explanatory variable	$\Delta \text{Premium}_t$		$\text{Unexpected issuance}_t$	
	Coeff	<i>t</i> -stat	Coeff	<i>t</i> -stat
<i>Intercept</i>	−0.017	−2.68**	−0.019	−0.54
$\Delta \text{premium}_{t-1}$	−0.245	−6.47**	0.496	2.56**
$\Delta \text{premium}_{t-2}$	−0.206	−5.04**	−0.074	−0.30
$\text{Unexpected issuance}_{t-1}$	−0.011	−2.34**	−0.308	−3.68**
$\text{Unexpected issuance}_{t-2}$	−0.011	−1.74*	−0.424	−5.42**
Adj. R^2		0.0992		0.2364
<i>N</i>		940		940

B. Convenience premium results with interaction terms

Explanatory variable	$\Delta \text{Premium}_t$		$\text{Unexpected issuance}_t$	
	Coeff	<i>t</i> -stat	Coeff	<i>t</i> -stat
<i>Intercept</i>	−0.017	−2.70**	−0.018	−0.53
$\Delta \text{Premium}_{t-1}$	−0.247	−6.53**	0.221	1.34
$\Delta \text{Premium}_{t-2}$	−0.207	−5.07**	0.016	0.06
$\text{Unexpected issuance}_{t-1}$	−0.022	−3.29**	−0.303	−3.65**
$\text{Unexpected issuance}_{t-2}$	−0.007	−0.79	−0.425	−5.48**
$\Delta \text{Premium}_{t-1} \times I_{\tau_S > 0}$			1.465	1.93*
$\Delta \text{Premium}_{t-2} \times I_{\tau_S > 0}$			−0.447	−0.68
$\text{Unexpected issuance}_{t-1} \times I_{\tau_S > 0}$	0.020	2.32**		
$\text{Unexpected issuance}_{t-2} \times I_{\tau_S > 0}$	−0.006	−0.49		
Adj. R^2		0.1019		0.2463
<i>N</i>		940		940

Panel A reports the results from the panel regressions of quarterly changes in the convenience premia for each state on quarterly bond issuance in excess of the average issuance over the prior three quarters for the corresponding state. We designate the latter as unexpected issuance. The dependent variable in the first column

Figure 8: Table 12

6 Short summary

- The paper studies whether tax-exempt municipal bonds contain a convenience premium for tax avoidance.
- It controls directly for credit risk using state CDS and for liquidity using taxable municipal bonds.
- The average convenience premium is 15.49 basis points.
- Premia are related to other convenience premia, tax uncertainty, state fund flows, migration, tax-minimizing behavior, and the TCJA SALT limitation.
- Term-structure tests reject the view that the measured premium is entirely a tax-rate risk premium.
- Issuers appear to respond to premia by increasing issuance.

7 Conclusion

7.1 Core conclusion

Fleckenstein and Longstaff show that state-issued tax-exempt municipal bonds contain significant convenience premia after controlling for credit risk, liquidity, and the maximum possible mechanical tax effect.

7.2 Economic intuition

Investors dislike taxes and actively use tax-advantaged vehicles to protect wealth. Tax-exempt munis provide a direct way to avoid taxation, so investors bid up their prices.

7.3 Takeaway

Convenience premia can arise not only from liquidity and safety, but also from freedom from taxation. This creates tax-related seigniorage for municipal issuers and has implications for the cost of capital in taxable debt markets.